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Slidable current collector and method for contacting conductor rail

Abstract

A slidable current collector has an array of terminals with carbon brushes for contacting conductor rails to deliver electrical power to a moving work machine. The terminals have upper sections with a conductive post, lower sections that include a reservoir of liquid metal, and bladders that connect the upper sections with the lower sections. Magnets surround outer shells of the terminals. Fluid above a threshold pressure fed into the bladders holds the upper sections apart from the lower sections and forces the magnets away from the conductor rails. Fluid below the threshold pressure allows the magnets to clamp the terminals to the conductor, lowers the conductive post into the liquid metal, and urges the carbon brushes against the conductor rails. The bladders provide a fluid suspension distributed across the array of terminals, enabling consistent electrical contact and wear for the carbon brushes.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to a slidable current collector and a method for contacting a conductor rail. More specifically, the present disclosure relates to a slidable current collector having a fluid suspension distributed across an array of terminals and a method for contacting the current collector on a conductor rail with multiple degrees of freedom while sliding.
BACKGROUND
(2) Heavy work machines, such as earth-moving vehicles or hauling trucks, require significant power to carry out their functions. The machines themselves can be of substantial weight, and their loads require large amounts of power to move. Diesel engines typically provide that power, but the use of machines powered by diesel engines may not be appropriate in certain environments. For instance, in some implementations, heavy work machines may need to travel large distances through rugged terrain. At a remote mining site, for example, groups of these machines are often employed to ferry extreme loads along roadways, or haul routes, extending between various locations within the mining site. Supplies of diesel fuel may be far away from such locations or not easily delivered to such locations. In addition, the groups of diesel machines can generate significant pollution.
(3) A power rail based on the ground may provide electrical power to traveling vehicles such as heavy work machines. In some environments, such as with trains or subways that travel on a fixed track, precise alignment between the fixed track and the power rail can ensure reliable delivery of electrical power to a current collector as the vehicle moves. For a heavy work machine that is freely steerable, however, establishing and maintaining an electrical connection between a current collector attached to an extended arm and the power rail can be particularly challenging. The rails may be slightly uneven, twisted, or curved, possibly leading to disconnections or arcing. Arcing can degrade current flow and damage components.
(4) In some environments, such as a mining site, the terrain can also interfere with continuous connection with power rails for a freely steerable work machine along a haul route. The haul route may be uneven, hilly, and pocked, which can lead to steering deviations that could also cause arcing at the current collector. These variations in terrain can also cause the machine to disconnect from the rail, detracting from the value of rail-based delivery of electrical power. While increasing adhesion between the current collector and the rail may decrease disconnections, increased adhesion at the current collector leads to unwanted drag on the arm of the work machine and accelerates wear on the current collector.
(5) One approach for contacting a current collector and a power rail is described in U.S. Pat. No. 3,804,997 (“the '997 patent”). The '997 patent describes a system for a train traveling at high speeds

that purports to counteract the fluttering of contact shoes sliding against a powered rail, which can interrupt electrical contact. A contactless force field, which may be magnetic or pneumatic, provides offsetting forces to stabilize the collector shoe against the power rail with fixed spacing during high-speed travel. The system of the '997 patent, however, addresses only fluttering of collector shoes with a vehicle traveling in fixed relation to a powered rail and does not contemplate positional deviations in multiple dimensions between a freely steerable vehicle and the powered rail. Nor does the system of the '997 patent address a risk of the collector shoes becoming detached from the power rail during the positional deviations. As a result, the system of the '997 patent is insufficient for freely steerable vehicles having current collectors sliding over a power rail along a haul route that may vary over diverse terrain.
(6) Examples of the present disclosure are directed to overcoming deficiencies of such systems.
SUMMARY
(7) In an aspect of the present disclosure, a terminal for an electric current collector includes a lower section, an upper section, and a bridging section. The lower section includes a carbon brush, a reservoir, and a conductive fluid within the reservoir. The reservoir and the conductive fluid are disposed about a central axis of the carbon brush, and the carbon brush is electrically connected with the conductive fluid. The upper section includes a post that is hollow and defines at least an upper chamber along the central axis. The bridging section includes a bladder and an insulative fluid within the bladder. The bladder is pliable, connects the lower section with the upper section, and extends radially around the central axis.
(8) In another aspect of the present disclosure, an electric current collector for sliding in a direction along a power rail includes a frame having a substrate with a top surface and an underside, a busbar positioned on the top surface, and a first terminal and a second terminal. The first terminal and the second terminal are aligned in a column along the direction and mounted between the top surface and the underside of the substrate. The busbar electrically connects the first terminal and the second terminal. The first terminal and the second terminal respectively include a carbon brush, a piston, a reservoir, and a conductive fluid within the reservoir. The carbon brush is aligned and electrically connected with the piston along a central axis, and the conductive fluid radially surrounds and contacts at least a portion of the piston along the central axis. The first terminal and the second terminal respectively also include an upper section having a hollow post that defines at least an upper chamber configured to loosely receive the piston along the central axis. The first terminal and the second terminal respectively also include a fluid suspension of an insulative fluid within a pliable bladder. The fluid suspension connects the reservoir with the post and enables movement of the carbon brush along the central axis.
(9) In yet another aspect of the present disclosure, a method includes touching an electrical contact of a slidable terminal on a rail surface, clamping the slidable terminal to the rail surface using, at least in part, an attractive force between a magnet surrounding a body of the slidable terminal and the rail surface, and applying a downward force along a central axis between the electrical contact and the rail surface. Applying the downward force includes directing a fluid through an upper section of the slidable terminal and into a region confined by an airbag and adjusting pressure for the fluid to a first pressure level within the region. The airbag couples the upper section and a lower section of the slidable terminal and is positioned radially around the central axis. The method further includes sliding the electrical contact across the rail surface.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is an isometric view of an electrically powered work machine coupled to a roadside power source in accordance with an example of the present disclosure.

(2) FIG. **2** is a partial isometric rear view of a conductive rod and trailing arms in accordance with an example of the present disclosure.
(3) FIG. **3** is an isometric view of a current collector having a plurality of terminals in accordance with an example of the present disclosure.
(4) FIG. **4** is an isometric view of a representative terminal and rail in accordance with an example of the present disclosure.
(5) FIG. **5** is a cross-sectional side view of the representative terminal of FIG. **4** in a state of full extension in accordance with an example of the present disclosure.
(6) FIG. **6** is a cross-sectional side view of the representative terminal of FIG. **4** in a state of partial compression in accordance with an example of the present disclosure.
(7) FIG. **7** is an isometric view of one carbon-brush assembly in accordance with an example of the present disclosure.
(8) FIG. **8** is a rear view of the current collector in FIG. **3** engaged on elevated power rails in accordance with an example of the present disclosure.
(9) FIG. **9** is a rear view of the current collector in FIG. **3** detached from elevated power rails in accordance with an example of the present disclosure.
(10) FIG. **10** is a flowchart of a method of engaging a slidable electrical contact with a conductor rail in accordance with an example of the present disclosure.

DETAILED DESCRIPTION

(11) Wherever possible, the same reference numbers will be used throughout the drawings to refer to same or like parts. Multiple instances of like parts within a figure may be distinguished using a letter suffix. FIG. **1** illustrates an isometric view of a work machine **100** within an XYZ coordinate system as one example suitable for carrying out the principles discussed in the present disclosure. Exemplary work machine **100** travels along a defined path or roadway, typically from a source to a destination within a worksite. In one implementation as illustrated, work machine **100** is a hauling machine that carries a load within or from a worksite within a mining operation. For instance, work machine **100** may haul excavated ore or other earthen materials from an excavation area along roads to dump sites and then return to the excavation area. In this arrangement, work machine **100** may be one of many similar machines configured to ferry earthen material in a trolley arrangement. While illustrated as a large mining truck in this instance, work machine **100** may be any machine that carries a load between different locations within a worksite, examples of which include Wherever possible, the same reference numbers will be used throughout the drawings to refer to same or like parts. FIG. **1** illustrates an isometric view of a work machine **100** within an XYZ coordinate system as one example suitable for carrying out the principles discussed in the present disclosure. Exemplary work machine **100** travels along a defined path or roadway, also termed haul route **101**, typically from a source to a destination within a worksite. In one implementation as illustrated, work machine **100** is a hauling machine that carries a load within or from a worksite within a mining operation. For instance, work machine **100** may haul excavated ore or other earthen materials from an excavation area along haul route **101** to dump sites and then return to the excavation area. In this arrangement, work machine **100** may be one of many similar machines configured to ferry earthen material in a trolley arrangement. While illustrated as a large mining truck in this instance, work machine **100** may be any machine that carries a load between different locations within a worksite, examples of which include an articulated truck, an off-highway truck, an on-highway dump truck, a wheel tractor scraper, or any other similar machine. Alternatively, work machine **100** may be an off-highway truck, on-highway truck, a dump truck, an articulated truck, a loader, an excavator, a pipe layer, or a motor grader. In other implementations, work machine **100** need not haul a load and may be any movable machine associated with various industrial applications including, but not limited to, mining, agriculture, forestry, construction, and other industrial applications.
(12) Referring to FIG. **1**, an example work machine **100** includes a frame **103** powered by electric

engine **102** to cause rotation of traction devices **104**. Traction devices **104** are typically four or more wheels with tires, although tracks or other mechanisms for engagement with the ground along haul route **101** are possible. Electric engine **102** functions to provide mechanical energy to work machine **100** based on an external electrical power source, such as described in further detail below. A primary example of mechanical energy provided by electric engine **102** is propelling traction devices **104** to cause movement of work machine **100** along haul route **101**, but electric engine **102** also includes components sufficient to power other affiliated operations within work machine **100**. For instance, in some implementations, electric engine **102** includes equipment for converting electrical energy to provide pneumatic or hydraulic actions within work machine **100**. While electric engine **102** is configured to operate from an external electrical power source, electric engine **102** typically includes one or more batteries for storing electrical energy for auxiliary or backup operations.

(13) In accordance with the principles of the present disclosure, work machine **100** further includes a conductor rod **106** configured to receive electrical power from a power rail **108**. In some examples, power rail **108** includes one or more beams of metal arranged substantially parallel to and at a distance above the ground. Support mechanisms hold power rail **108** in place along a distance at the side of a haul route **101** for work machine **100** to traverse. The support mechanisms and power rail **108** may be modular in construction, enabling their disassembly and reassembly at different locations or their repositioning along the existing haul route **101**. Moreover, while shown in FIG. **1** to the left of work machine **100** (along the Y axis) as work machine **100** travels in the direction of the X axis, power rail **108** may be installed to the right of work machine **100** (along the −Y axis) or in other locations suitable to the particular implementation. In many examples, such as within a mining site, power rail **108** will not be configured continuously at a fixed distance along a side of haul route **101** and at a fixed height above the ground due, at least in part, to the variation of the terrain. Therefore, it is expected that the vertical, horizontal, and angular positions of the surface of power rail **108** in the XYZ planes will vary along haul route **101**.

(14) Power rail **108** provides a source of electrical power for work machine **100** as either AC or DC voltage. In some examples, power rail **108** has two or more conductors, each providing voltage and current at a different electrical pole. In one implementation (e.g., an implementation in which the power rail **108** includes three conductors), one conductor provides positive DC voltage, a second conductor provides negative DC voltage, and a third conductor provides a reference voltage of 0 volts, with the two powered conductors providing +1500 VDC and −1500 VDC. These values are exemplary, and other physical and electrical configurations for power rail **108** are available and within the knowledge of those of ordinary skill in the art.

(15) Conductor rod **106** enables electrical connection between work machine **100** and power rail **108**, including during movement of work machine **100** along haul route **101**. In the example shown in FIG. **1**, conductor rod **106** is an elongated arm resembling a rigid pole. FIG. **1** shows conductor rod **106** positioned along a front side of work machine **100**, with respect to the direction of travel of work machine **100** in the direction of the X axis. In this arrangement, conductor rod **106** is located in FIG. **1** in the Y-Z plane essentially along the Y axis with a proximal end near a right side of work machine **100** and a distal end at a left side of work machine **100**. Conductor rod **106** may be attached to any convenient location within work machine **100**, such as to frame **103**, in a manner to enable conductor rod **106** to reach and couple to power rail **108**. Shown in FIG. **1** as extending to a left side of work machine **100** toward power rail **108**, conductor rod **106** may alternatively be arranged to extend to a right side (along the −Y axis) and at any desired angle from work machine **100** such that conductor rod **106** may be coupled to power rail **108** for obtaining electrical power.

(16) As embodied in FIG. **1**, conductor rod **106** includes a cylinder portion **109** mounted to frame **103** of work machine **100**. Cylinder portion **109** has a hollow interior and may be a conductive metal having suitable mechanical strength and resiliency, such as aluminum. Within cylinder portion **109**, an extension **110** is retained. Extension **110** is slidably engaged within cylinder portion

109 of conductor rod **106** such that it may be extended or retracted axially, i.e., along the Y axis in FIG. **1**, to adjust the reach of conductor rod **106**. Specifically, in a retracted position, extension **110** is caused to slide within cylinder portion **109** of conductor rod **106** such that a length of conductor rod **106** roughly spans the width of work machine **100**. A junction **112** serves as the interface between extension **110** and cylinder portion **109**, which is the main body of conductor rod **106**. When extension **110** is fully retracted or collapsed into cylinder portion **109**, junction **112** essentially becomes the left edge of conductor rod **106**. On the other hand, when extension **110** is extended from cylinder portion **109** of conductor rod **106**, extension **110** may reach from work machine **100** to above or near power rail **108** on the side of haul route **101**.

(17) Within, and possibly including cylinder portion **109**, conductor rod **106** has a series of electrical conductors passing longitudinally, i.e. along the Y axis in FIG. **1**, at least from a base **122** at a proximal end to a tip **124** at a distal end. Typically, the conductors within conductor rod **106** are formed of a metallic material. Moreover, as with cylinder portion **109**, the material for conductors within conductor rod **106** typically have suitable mechanical strength and resiliency to permit their stable extension from work machine **100** to above power rail **108** at the side of haul route **101**. In some examples, the conductors are concentric tubes, or hollow cylinders, of solid metal such as aluminum nested together and sized to provide electrical capacity sufficient for powering work machine **100**. Tubular conductors within extension **110** slidably engage with corresponding tubular conductors in the portion of conductor rod **106** mounted on work machine **100**. This engagement while the tubes slide ensures electrical continuity during extension or retraction of conductor rod **106**.

(18) At tip **124** of extension **110** within conductor rod **106**, a connector assembly **114** provides an interface to power rail **108** via trailing arms **116** and current collector **118**. The arrangement of connector assembly **114**, trailing arms **116**, and current collector **118** of FIG. **1**, which are collectively also referred to as a trailing assembly, are described in further detail in FIG. **2**. Power rail **108** is typically arranged along a side of haul route **101**, and work machine **100** traverses haul route **101** substantially in parallel with power rail **108**. Thus, in reference to FIG. **1**, power rails **108** and a travel path for work machine **100** are substantially in parallel with each other and with the X axis. Current collector **118** is configured to maintain an electrical connection with power rail **108** while sliding along its surface in the direction of the X axis as work machine **100** moves. In some examples, trailing arms **116** are conductors coupled to current collector **118**, each conducting voltage and current at a different electrical pole for respective conductors within conductor rod **106**. In operation, electrical power is accessed from power rail **108** via current collector **118**, which remains in contact during movement of work machine **100**, and the electrical power is conducted through trailing arms **116** into connector assembly **114**.

(19) From connector assembly **114**, the electrical power is conveyed at tip **124** through the nested tubular conductors within extension **110** and cylinder portion **109** to head **122** of conductor rod **106** and through a head-end interface **120** to work machine **100**. Head-end interface **120** provides at least an electrical connection between conductor rod **106** and work machine **100** for powering electric engine **102** and otherwise enabling operations within work machine **100**. In some examples, head-end interface **120** may also provide an interface for controls between work machine **100** and conductor rod **106**. In some examples, head-end interface **120** includes passageways to control mechanical operation of conductor rod **106**, such as for pressurized fluid of a pneumatic or hydraulic control system to extend and retract extension **110** or to control operations within current collector **118** in a manner described further below. In other examples, head-end interface **120** includes passageways for signals to communicate with electronic controls.

(20) Connector assembly **114** not only provides electrical connection between the conductors within extension **110** of conductor rod **106** and trailing arms **116**, but also accommodates the various changes in relative position between power rail **108** and work machine **100** during travel along haul route **101**. Those changes in relative position can include multiple deviations, such as

those occurring laterally (work machine **100** and connector assembly **114** moving in the Y axis relative to current collector **118**), vertically (work machine **100** and connector assembly **114** moving in the Z axis relative to current collector **118**), and angularly (work machine **100** and connector assembly **114** moving in the X-Y plane angularly around the Z axis). One or all of these deviations could occur as a driver steers work machine **100** along haul route **101**, work machine **100** responds to an uneven or pocked roadway, or an orientation of power rail **108** varies with respect to work machine **100**. FIG. **2** is discussed below and illustrates an example trailing assembly **200** suitable for accommodating multiple deviations in position between work machine **100** and current collector **118**.

(21) FIG. **2** is a view from a side of power rail **108** opposite work machine **100** facing generally forward (i.e., along the X axis), which shows a trailing assembly **200** from a back side of extension **110**. As shown in the example of FIG. **2**, power rail **108** contains three conductors separately identified as inner rail **108**A, middle rail **108**B, and outer rail **108**C, with “inner,” “middle,” and “outer” describing a respective position relative to work machine **100**. As well, in this instance and elsewhere in this disclosure, the suffix “A” denotes a component associated with an electrical path including inner rail **108**A, suffix “B” denotes a component associated with an electrical path including middle rail **108**B, and suffix “C” denotes a component associated with an electrical path including outer rail **108**C. The absence of an “A,” “B,” or “C” suffix may connote a component discussed generically or collectively in the system without association with a specific power rail.

(22) In one example, two of the conductors in FIG. **2** provide electrical power at different polarities while the third conductor provides a reference of 0 volts. In other examples, the conductors can provide AC voltage at three different polarities or power rail **108** and conductor rod **106** can contain fewer or more than three conductors. Current collector **118**, described in more detail below, is electrically coupled to power rail **108** and slides along its surface to maintain an electrical connection with each of inner rail **108**A, middle rail **108**B, and outer rail **108**C. Ultimately, current collector **118** provides the electrical interface between power rail **108** and trailing arms **116**.

(23) As shown in FIG. **2**, connector assembly **114** within trailing assembly **200** is integrated into extension **110**. Although not shown, connector assembly **114** bridges between three conductors within extension **110** and a rotational interface **202**, which exits at a bottom portion of extension **110**. Rotational interface **202** in turn couples the three conductors within **110** to respective trailing arms **116**, namely, outer arm **210**, inner arm **212**, and middle arm **214**, and uses socket and hinge joints to enable ranges of motion for trailing arms **116**. As well, rotational interface **202** permits twisting of trailing arms **116** about an axis A-A. Referring to outer arm **210**, rotational interface **202** connects to a first conductor (not shown) within extension **110** and includes outer lug **204** arranged generally parallel to the Y axis. A socket joint within outer lug **204** provides rotational movement for outer arm **210** about the axis B-B. In the illustrated example, an outer hinge **228** is coupled to outer lug **204** and further enables outer arm **210** to rotate about an axis C-C, as shown in FIG. **2**.

(24) Referring to inner arm **212**, rotational interface **202** connects to a second conductor (not shown) within extension **110** and includes an inner lug **206** that provides a rotational movement about axis B-B within a socket joint similar to outer lug **204**. As well, inner lug **206** is coupled to an inner hinge **232**, which enables inner arm **212** to pivot about an axis D-D, as shown in FIG. **2**. A similar socket and hinge arrangement is illustrated in FIG. **2** for coupling middle arm **214** to a third conductor (not shown) within extension **110**. In particular, a middle lug **208** exits rotational interface **202** and provides rotational ability of middle arm **214** about a central axis (not labeled) perpendicular to axis B-B. Further, a middle hinge **230** is coupled to middle lug **208** and provides freedom for second arm **214** to pivot about an axis (not shown) parallel to axis B-B. In short, each of trailing arms **116** is coupled to extension **110** via a pivoted and hinged interface that permits movement of the arms with multiple degrees of freedom with respect to extension **110**.

(25) At their opposite ends, trailing arms **116** connect via similar configurations with current collector **118** allowing movement of the arms with multiple degrees of freedom with respect to

current collector **118**. In some examples, the components within **118** associated with trailing arms **116** have axes of rotation parallel to those within connector assembly **114**, such as about axes A-A, B-B, C-C, and D-D, but the additional axes within current collector **118** are not shown on FIG. **2** for simplicity. In some examples, for outer rail **108**C, outer arm **210** is connected to an outer contactor hinge **216** and an outer contactor lug **218**. For inner rail **108**A, inner arm **212** is connected to an inner contactor hinge **220** and an inner contactor lug **222**. For middle rail **108**B, middle arm **214** is connected to a middle contactor hinge **224** and a middle contactor lug **226**. These mechanical linkages are illustrative only and are described to provide context for the discussion of movement by current collector **118** below.

(26) Electrically, the connections within trailing assembly **200** provide a path for the conduction of electrical power from power rail **108** to the conductors within extension **110**. As work machine **100** moves along haul route **101**, extension **110** pulls trailing arms **116** generally parallel to the X axis. Current collector **118** rests on power rail **108** and is thereby caused to slide along the surface of inner rail **108**A, middle rail **108**B, and outer rail **108**C during movement of work machine **100**. While sliding, current collector **118** maintains physical and electrical contact with power rail **108**, passing electrical power from power rail **108** to trailing arms **116**.

(27) Mechanically, the combined actions of the lugs and hinges within rotational interface **202** and within current collector **118** enable flexible and versatile movement of trailing arms **116** between extension **110** and current collector **118**. As work machine **100** moves forward (in the direction of the X axis), trailing arms **116** and current collector **118** trail behind extension **110**. If the position of extension **110** changes with respect to the position of current collector **118**, trailing assembly **200** permits trailing arms **116** to move in several directions and maintain the connection between current collector **118** and power rail **108**. Risks of current collector **118** disconnecting from power rail **108** or causing arcing due to smaller deviations as work machine **100** travels along haul route **101** may be caused by variations in the alignment of power rail **108** or imperfections in the surface conditions of inner rail **108**A, middle rail **108**B, or outer rail **108**C. FIGS. **3-10** illustrate features of current collector **118** to provide a secure and quality electrical contact with power rail **108** despite deviations at the rails.

(28) FIG. **3** illustrates an isometric view of current collector **118** looking outward and backward with respect to conductor rod **106**, i.e., looking between the −X axis and the +Y axis. In other words, FIG. **3** essentially shows a view of current collector **118** from extension **110** as current collector **118** rests on power rail **108**. In the illustrated examples, current collector **118** generally includes at least a frame **302**, busbars (referred to collectively as **304**) as conductors associated with respective trailing arms **116**, and one or more terminals (referred to collectively as **306**) interfacing between power rail **108** and busbars **304**. Each of these components is discussed below.

(29) Frame **302** serves as a structural base for current collector **118**. To decrease drag on conductor rod **106**, frame **302** may be constructed with dielectric materials that are light in weight yet structurally resilient. In some examples, an exterior of frame **302** is made of fiberglass while an interior is a foamed material such as a polymer, and the overall frame weighs about 5 KG. Other material options and weights for frame **302** are available and will be apparent to those of ordinary skill in the art based on the intended implementation. As illustrated in FIG. **3**, frame **302** roughly resembles a hat in its shape and may have a substrate **308** within the X-Y plane at its center. Substrate **308** is substantially flat across the X-Y plane and configured in operation to be positioned on top of and parallel to power rail **108**. At opposing lateral sides (i.e., along the Y axis), substrate **308** merges with an angled inner side **310** and an angled outer side **312**, which merge respectively with an inner flange **314** and an outer flange **316**. The angled inner side **310** and angled outer side **312** each transition a height of frame **302** (dimensionally along the Z axis) from a top surface **322** on substrate **308** to inner flange **314** and outer flange **316**. An inner bumper **318** and an outer bumper **320** are attached to frame **302** at an underside **324** of substrate **308**. Variations in the dimensional characteristics of frame **302**, particularly a width along the X axis for substrate **308**

and a slope of angled inner side **310** and angled outer side **312**, together with the role of inner bumper **318** and outer bumper **320** are discussed with respect to FIGS. **7** and **8** below.
(30) Mounted on top surface **322** of substrate **308**, busbars **304** are conductors that provide electrical interconnections between current collector **118** and trailing arms **116**. Accordingly, busbars **304** in some examples are plates of aluminum or similar conductive material attached to top surface **322** through bolts or similar attachment devices. One busbar is shown for each rail within power rail **108**. For a system as in FIGS. **1** and **2** having three power rails, substrate **308** includes an inner busbar **304**A corresponding to inner rail **108**A, a middle busbar **304**B corresponding to middle rail **108**B, and an outer busbar **304**C corresponding to outer rail **108**C. In some examples, each of the busbars **304** is secured on top surface **322** as rectangular strips positioned in general alignment with its corresponding power rail. Thus, in the example of FIG. **3**, inner busbar **304**A is mounted on substrate **308** parallel to the X axis and to inner rail **108**A underneath (FIG. **2**), as is middle busbar **304**B parallel to and over middle rail **108**B and **304**C over outer rail **108**C. Each of the busbars **304** provides a connection to a corresponding one of trailing arms **116** (FIG. **2**). Specifically, inner contactor lug **222** serves as an electrical and mechanical interface between inner busbar **304**A and inner arm **212** (FIG. **2**), as does middle contactor lug **226** for middle busbar **304**B and middle arm **214** and outer contactor lug **218** for outer busbar **304**C and outer arm **210**.
(31) In addition, current collector **118** includes one or more terminals **306** that are secured within each of the busbars **304** and substrate **308**. Described in more detail below with respect to FIGS. **4-6**, terminals **306** provide electrical conductivity between power rail **108** and busbars **304**. Specifically, while not shown in FIG. **3**, each of inner busbar **304**A, middle busbar **304**B, and **304**C and substrate **308** have holes through them for receiving and retaining terminal **306**. At underside **324** of substrate **308**, terminals **306** contact power rail **108** and remain in contact as work machine **100** travels along haul route **101**. Terminals **306** conduct electrical power from power rail **108** to busbars **304**, which then conduct the electrical power to trailing arms **116** for use by work machine **100**. In some examples, a plurality of terminals **306** are employed for each power rail. In the example of FIG. **3**, current collector **118** has three terminals **306** accessing electrical power from each power rail. Namely, first inner terminal **306**A-**1**, second inner terminal **306**A-**2**, and third inner terminal **306**A-**3** are affixed within inner busbar **304**A and configured to slide across inner rail **108**A. In this context, the suffix “-n” designates multiple occurrences of a component associated with a common conductor rail, e.g., “A-1” and “A-2” as multiple ones of the same component associated with inner rail **108**A. Similarly, first middle terminal **306**B-**1**, second middle terminal **306**B-**2**, and third middle terminal **306**B-**3** are affixed within middle busbar **304**B and configured to slide across middle rail **108**B. Finally, first outer terminal **306**C-**1**, second outer terminal **306**C-**2**, and third outer terminal **306**C-**3** are affixed within **304**C and configured to slide across outer rail **108**C.
(32) Above top surface **322** of substrate **308**, only a portion of terminals **306** is visible. Illustrated in the lower left corner of FIG. **3** for first inner terminal **306**A-**1**, a first inner shoulder **326**A-**1** encircles a portion of first inner terminal **306**A-**1** above inner busbar **304**A and provides electrical contact between inner rail **108**A and inner busbar **304**A. Above first inner shoulder **326**A-**1**, a first inner neck **328**A-**1** and a first inner cap **330**A-**1** extends at the uppermost portion of first inner terminal **306**A-**1**. First inner neck **328**A-**1** provides additional electrical conductivity between inner rail **108**A and inner busbar **304**A. First inner shoulder **326**A-**1**, first inner neck **328**A-**1**, and first inner cap **330**A-**1** are discussed below with respect to FIGS. **4-6**. Similar structures may exist for the other terminals **306** in current collector **118**.
(33) Although FIG. **3** depicts redundancy of terminals **306** for each power rail, more or fewer than three terminals (including one) may be used for each rail. The redundancy can increase the quality and continuity of contact with a respective power rail, guard against degradation in the contact due to wear or operation of an individual terminal, and distribute mechanical and electrical loads across

more than one terminal. The array of terminals **306**, shown in FIG. **3** as a matrix of three terminals **306** for each of three rails in an arrangement of rows and columns, can be networked together to improve their performance. In a manner discussed in more detail below with respect to FIGS. **4-6**, pressurized fluid may enhance the operation of one or more of terminals **306**. To that end, tubes **332** feed pressurized fluid, typically air, across the array of terminals **306**. While not shown, tubes **332** may also pass from conductor rod **106**, through trailing arms **116**, and into one or more of inner contactor lug **222**, middle contactor lug **226**, or outer contactor lug **218**. A pneumatic or hydraulic control system within conductor rod **106** and/or work machine **100** can regulate and deliver the pressurized fluid through passageways in trailing arms **116**, through tubes **332**, and into caps **330** for each of the terminal **306**. While FIG. **3** provides an overview of the general components on current collector **118**, FIGS. **4-6** isolate a representative terminal within current collector **118** to show its structure and function.

(34) FIG. **4** illustrates a representative terminal **306** separated from current collector **118** and positioned on a representative one of the power rails. The figure shows a terminal **306** from a perspective looking toward work machine **100** as terminal **306** would slide to the left (parallel to the X axis) during forward movement of work machine **100**. As terminal **306** is representative, reference numbers for its parts do not include suffixes tied to particular power rails or locations within a frame **302**, and the structure and operation of terminal **306** could be applied to one or more other contactor terminals within current collector **118**. Similar to what is shown in FIG. **3**, shoulder **326**, neck **328**, and cap **330** make up an external upper portion of terminal **306**, which is exposed above top surface **322** in current collector **118**. Cap **330** includes an passageway **402**, which is a central orifice passing into terminal **306**. As shown in FIG. **3**, tubes **332** made of elastomeric or similar material may connect with cap **330** and feed pressurized fluid, such as air or other gas, into passageway **402**. In some examples, cap **330** is a so-called quick-connect device, providing the ability to be snapped onto or removed from terminal **306** with slight manipulation and force applied along the Z axis. In other examples, cap **330** is threaded and can be screwed on or unscrewed from terminal **306** as needed. Shoulder **326** and neck **328** are an upper part of a body **404** that serves as an exterior body for terminal **306**. As shown, in one example, body **404** is a substantially cylindrical shape and fits snugly within a similarly sized orifice through a corresponding one of the busbars **304** and through substrate **308**. Body **404** may be made of any conductive and resilient material, such as an aluminum alloy or similar material.

(35) At a lower portion of terminal **306**, a slidable electrical contact labeled as carbon brush **406** extends from a bottom of body **404**. While described as a carbon brush, the element at **406** may be any slidable electrical contact known in the art under different terms, such as shoes, brushes, wipers, etc., that interface with and enable electrical conduction from power rails **108** into terminals **306**. Specifically, carbon brush **406** contacts one of power rails **108** and initiates current transfer from power rail **108** for delivery to work machine **100**. Carbon brush **406** is typically composed of graphite mixed with one or more metals, such as copper, silver, or a copper alloy. In some examples, carbon brush **406** is substantially solid and generally has a disk-shape, similar to a hockey puck. A circular lower surface of carbon brush **406** contacts rail surface **408** and is sized to conduct amperage sufficient for the implementation. In one example, carbon brush **406** should transmit up to about 1000 Amps and is accordingly sized with an effective contact area of about six square inches. Other sizes for carbon brush **406** and current values are possible and within the knowledge of those of ordinary skill in the art.

(36) Above carbon brush **406**, a magnet **410** above a sleeve **412** encircles body **404**. In some examples, magnet **410** is a permanent magnet with poles arranged overall so that a magnetic force pulls magnet **410** toward power rail **108**. In one implementation, magnet **410** is a samarium cobalt magnet having an array configuration with about 68 lb-ft attraction force at a ¼ inch gap. These values are representative only and can vary without departing from the principles of the present disclosure. In other examples, magnet **410** may be an electromagnet with electrical configurations

provided to generate a similar magnetic force. In FIG. **4**, magnet **410** is shown elevated in the Z axis above power rail **108** to reveal carbon brush **406**, which is a position for magnet **410** during initial attachment and detachment with power rail **108**. In some examples, a rail surface **408** is positioned as a top layer over each of the power rails **108** and may be magnetic stainless steel or a similar material. After placing carbon brush **406** on rail surface **408**, magnet **410** is pulled downward toward rail surface **408**, as explained below with respect to FIGS. **5** and **6**. The magnetic attraction between magnet **410** and rail surface **408** provides a secure attachment between terminal **306** and power rail **108** during operation of work machine **100**. As illustrated, sleeve **412** is positioned below magnet **410**. Sleeve **412** may be made of plastic having a low coefficient of friction, which can enhance sliding between sleeve **412** and rail surface **408**.

(37) FIG. **5** depicts a cross-section of terminal **306** in FIG. **4**. The cross-section slices terminal **306** along the X-Z plane and shows the internal structure of terminal **306** looking in the −Y direction. Several features of terminal **306** discussed above for FIG. **4** are depicted in FIG. **5** along the exterior of terminal **306**, such as body **404**, magnet **410**, and cap **330**. The internal structure of terminal **306** shown in FIG. **5** generally separates into three sections: a lower section **500**, an upper section **520**, and a bridging section **540**.

(38) Within lower section **500**, carbon brush **406** described above provides an interface with rail surface **408** to conduct electrical energy into terminal **306**. Carbon brush **406** in the example illustrated is connected by rivets or other fastening devices to plate **502**. Plate **502** may be made of a conductive material such as aluminum or copper for providing a structurally resilient surface for securing carbon brush **406** in place. Plate **502** further secures a position for a piston **504**, which extends vertically along a central axis **510** upwardly in the direction of the Z axis away from carbon brush **406**. Piston **504** is likewise an electrically conductive material such as aluminum alloy or copper. Piston **504** in some examples is cylindrical in shape and is surrounded by reservoir **506**. As shown, reservoir **506** extends from or around plate **502** upwardly about a central axis **510**. Being an electrically conductive material with structural stability, such as an aluminum alloy, reservoir **506** in some examples has inner surface **508** along an inner circumference that provides an annular or bowl shape to reservoir **506** and defines a lower chamber **512** within terminal **306**. A conductive fluid **514** at least partially fills lower chamber **512** between piston **504** and inner surface **508**. Conductive fluid **514** may be any flowable conductive material. In one example, conductive fluid **514** is Galinstan, which is a eutectic alloy composed of gallium, indium, and tin. Galinstan melts at −19 C (−2 F) and is thus liquid at room temperature. Mercury or other liquid metals or conductive fluids having comparable electrical and rheological properties to Galinstan may be used and are considered to be within the scope of the presently disclosed subject matter. Bearings **516**, made of polymer or stainless-steel as an example, are positioned around piston **504** at a bottom (in the direction of the −Z axis) of lower chamber **512** and enable piston **504** and carbon brush **406** to twist angularly about central axis **510**. Further, a rotary seal **518**, adjacent bearings **516** and formed from Teflon as an example, prevents leakage of conductive fluid **514** from reservoir **506**.

(39) Within upper section **520** of terminal **306**, cap **330** connects to a post **522** to provide pressurized fluid into terminal **306** and to conduct electrical energy to neck **328**. FIG. **5** illustrates a slightly different and interchangeable configuration for cap **330** than in FIG. **4**. In particular, cap **330** in FIG. **4** is shaped as a quick-connect attachment that may be snapped in place within an upward opening of post **522**, while cap **330** in FIG. **4** has a threaded attachment. When in place on post **522**, cap **330** provides a fluid interface between passageway **402**, which passes within an interior space of cap **330** along central axis **510**, and an cavity **524**. Post **522** is an electrically conductive material having solid structural properties, such as an aluminum alloy, copper, or similar metals, and is configured to form two openings about central axis **510**: cavity **524** and upper chamber **526**. Cavity **524** is formed within an upper region of post **522** (in the direction of the Z axis) defined by first interior wall **523** concentrically surrounding central axis **510**. Post **522** is further configured to have a second interior wall **525** concentrically surrounding central axis **510** in

a lower region of post **522** that defines upper chamber **526**. Exterior to post **522**, optional strut **528** is positioned along a ledge of post **522** and serves as a brace to convey mechanical forces between post **522** and shoulder **326**. As such, shoulder **326** rests on and may press downwardly against strut **528** due to magnetic forces pulling magnet **410** toward **408**, in a manner discussed further below. Conversely, strut **528** may press upwardly against shoulder **326** in response to forces from passageway **402** acting within cavity **524**, in a manner discussed below.

(40) Within bridging section **540** of representative terminal **306**, a rod **542** and a bladder **546** connect lower section **500** with upper section **520**. In some examples, rod **542** is a bolt or similar structure as shown in FIG. **5** that extends longitudinally along central axis **510** between piston **504** and cavity **524**. As a result, in the example of FIG. **5**, rod **542** passes at least in part within upper chamber **526**. At one end (farther down central axis **510** in the −Z direction), rod **542** is secured to piston **504**, such as through a threaded engagement, an adhesive, a weld, or other suitable means. At an opposite end (farther up central axis **510** the +Z direction), rod **542** may have a head **544**. Head **544** is typically a same material as rod **542**, such as metal, and has a width within the X-Y plane greater than rod **542** along central axis **510**. Rod **542** is loosely arranged within post **522** such that rod **542** may freely move with piston **504** at least along central axis **510**.

(41) Bladder **546** is a sleeve formed of compressible material such as one or more layers of rubber possibly having a nylon reinforcement. As with an airbag or air spring, bladder **546** may have a substantially tubular or bulbous shape and is formed around central axis **510**. In some examples, one end of bladder **546** is fixed in place at upper attachment **548** between sides of post **522** and strut **528**, as shown in FIG. **5**. At an opposite end, bladder **546** may be fixed in place to a radially outer side of reservoir **506** at lower attachment **550**. Bladder **546** is typically configured with some slack between upper attachment **548** and lower attachment **550**, as indicated by fold **552**, to accommodate movement upward and downward along central axis **510** by lower section **500** of terminal **306**. Bladder **546** defines middle chamber **554**, which is an open cavity within its walls sufficient to contain one or more flowable substances. Upper attachment **548** and lower attachment **550** provide a sealed connection between bladder **546**, post **522**, and reservoir **506** sufficient to contain the one or more flowable substances within middle chamber **554** without leakage. In some examples, the one or more flowable substances within middle chamber **554** may include an insulative fluid **556**, which may be any liquid or gas having electrical insulative properties, such as air. Insulative fluid **556** may be the same or a different substance as the pressurized fluid introduced into passageway **402**. In other examples, including as shown in FIG. **6** and discussed below, the one or more flowable substances within middle chamber **554** may include ones that are electrically conductive, such as conductive fluid **514**.

(42) Turning to operation of terminal **306**, FIGS. **4** and **5** illustrate a configuration in which terminal **306** is in the process of being attached or detached from a representative power rail **108**. In this configuration, pressurized fluid is used to help keep magnet **410** separated from rail surface **408** to provide maneuverability for carbon brush **406** and terminal **306** to be positioned, typically by hand. In some examples, attachment or detachment occurs by inserting a fluid such as air within passageway **402** at a pressure above atmospheric pressure sufficient to cause post **522** to be lifted axially in the direction of the Z axis. Introduced from one of tubes **332** (FIG. **3**), the pressurized fluid enters passageway **402** along central axis **510**, applying an expansive force indicated as F**1** at least within upper chamber **526**. Pressurized fluid in the arrangement of FIG. **5** will also enter middle chamber **554**, causing pliable bladder **546** to expand. As a result, expansive force F**1** will tend to push post **522** upwardly, which in turn will press strut **528** upwardly against shoulder **326**, and will push carbon brush **406** against rail surface **408** (downward force F**2** in FIG. **5**). On the other hand, magnet **410**, which may surround body **404** and rest on a lower lip of body **404** above **412**, has a downward gravitational pull due to its mass and, when terminal **306** is positioned on power rail **108**, magnetic attraction toward rail surface **408**. These combined gravitational and magnetic forces from magnet **410** are denoted as F**3** in FIG. **5**. In the state illustrated in FIG. **5**,

fluid introduced to passageway **402** is at sufficient pressure to generate upward force F**1** against shoulder **326** in excess of the downward magnet force F**3** pulling between magnet **410** and rail surface **408**. As a result, as shown in FIGS. **4** and **5**, with sufficiently high fluid pressure fed into passageway **402**, magnet **410** may be lifted, or otherwise kept apart, from rail surface **408**.
(43) In the configuration depicted in FIG. **5**, magnetic attraction between magnet **410** and rail surface **408** is counteracted, and an operator can readily attach or detach terminal **306** as part of current collector **118** with respect to power rail **108**. Moreover, with force F**2** exceeding force F**3**, post **522** will be lifted, or otherwise kept apart, from reservoir **506** and a conductive-fluid surface **560**, ensuring electrical isolation between lower section **500** and upper section **520**. In some examples, such as with magnet **410** being a samarium-cobalt magnet with about 68 lb-ft attraction force at ¼ inch gap, air or other fluid provided within passageway **402** above about 20 PSI can provide sufficient pneumatic or hydraulic force F**1** to overcome the downward pull F**3** on body **404**, resulting in the separated configuration for terminal **306** as shown in FIGS. **4** and **5**. These values are representative only and other pressure values may achieve similar results based on the components employed for an implementation. Variation and alternatives are within the knowledge of those of ordinary skill in the art.
(44) The axial separation between post **522** and reservoir **506** will generally be proportional to the fluid pressure maintained within upper chamber **526**. At one extreme, shown in FIG. **5**, the fluid pressure is sufficiently high to force post **522** upwardly along central axis **510** to a point that an upper stop **558** within post **522** contacts a lower surface of head **544**. This contact creates a blockage to halt further upward movement of post **522** along central axis **510** with even higher values of fluid pressure, such as above 20 PSI in some examples, fixing lower section **500** and upper section **520** of terminal **306** in a position of full extension or separation. With lower values of fluid pressure, such as below 20 PSI in some examples, the ratio between upward force F**1** and downward magnet force F**3** lessens, and lower section **500** and upper section **520** of terminal **306** will move closer to each other. FIG. **6** illustrates an example of terminal **306** in FIG. **5** after moving into a position of less than full extension.
(45) In FIG. **6**, fluid pressure fed into passageway **402** is less than in FIG. **5**, such as less than about 20 PSI in some examples. In this arrangement, the fluid pressure within upper chamber **526** is an amount such that expansive pneumatic or hydraulic force F**1** does not exceed the downward magnet force F**3**. As a result, as shown in FIG. **6**, magnet **410** pulls body **404** downward compared with its position in FIG. **5** and forces sleeve **412** into contact with rail surface **408**. Phantom lines in FIG. **6** show the difference in position for the exterior of terminal **306** in FIG. **5** compared with in FIG. **6**. Terminal **306** is thereby held in engagement with rail surface **408** sufficiently to resist external forces, such as those imparted by work machine **100** moving across rough surfaces of haul route **101**, that may urge separation of terminal **306** from rail surface **408**. On the other hand, attraction between magnet **410** and rail surface **408** is low enough to permit sleeve **412** to slide along rail surface **408** as current collector **118** is pulled by trailing arms **116** without causing undue drag. Selection of appropriate magnetic properties for magnet **410** based on the components chosen for a particular implementation will be within the knowledge of those of ordinary skill in the art.
(46) Referring to the interior of terminal **306** in the cross-section of FIG. **6**, the decrease in pneumatic or hydraulic force F**1** results in magnet force F**3** pulling shoulder **326**, strut **528**, and post **522** downward relative to rod **542**. As upper section **520** is lowered, a post bottom **602** dips into conductive fluid **514** within lower chamber **512** of reservoir **506**. Contact between the solid metal of post bottom **602** and conductive fluid **514**, which may be a liquid metal such as Galinstan, completes an electrical path from carbon brush **406** to neck **328**. To mitigate the risk of arcing, insulative fluid **556** may be partly or wholly oil or another inert media rather than air. In some examples, the inert media has the same or greater density than the pressurized fluid entering passageway **402** and a lower density than conductive fluid **514**, causing the inert media to cover conductive-fluid surface **560**. As post bottom **602** enters lower chamber **512** and displaces a portion

of conductive fluid **514**, conductive-fluid surface **560** will rise into middle chamber **554** within bladder **546**. It will be apparent that the amount of immersion for post **522** within conductive fluid **514** can be controlled based on the fluid pressure provided into passageway **402**. As noted for FIG. **5**, at high values of fluid pressure, such as above 20 PSI in some examples, no immersion for post **522** will occur and terminal **306** will be in a position of full extension. On the other hand, with upper chamber **524** at atmospheric pressure, post **522** may be lowered into immersion in conductive fluid **514** to an extent where post bottom **602** contacts a lower stop **604** on reservoir **506**, i.e., an extreme position of full compression. At pressure values within upper section **520** in between the extremes of full extension and full compression, such as about 5 PSI for some examples, post bottom **602** may be immersed within conductive fluid **514** without contacting lower stop **604**, as illustrated in FIG. **6**.

(47) Accordingly, the described arrangement enables terminal **306** to effectively function as a pneumatically or hydraulically controlled mechanical and electrical switch. Mechanically, magnet **410** and rail surface **408** can be forced apart and allowed to contact based on application of appropriate fluid pressure to terminal **306**. Electrically, a position of post **522** can be moved axially to be separated from conductive fluid **514** (open circuit) or immersed in conductive fluid **514** (closed circuit) using appropriate fluid pressure. The combination of mechanically forcing magnet **410** apart from rail surface **408** and forcing post **522** apart from conductive fluid **514** in full extension increases safety to personnel during attachment and detachment of current collector **118** to power rail **108** in the event power rail **108** is electrified. Moreover, the pneumatic or hydraulic networking of an array of terminals **306** within current collector **118** permits the control to affect all terminals **306** within the current collector. As such, current collector **118** can be commanded pneumatically or hydraulically from a source such as work machine **100** to open or close an electrical circuit within current collector **118** and to engage or disengage a magnet clamping force against the **108**.

(48) In addition to adjusting immersion or separation of post **522** with respect to lower attachment **550**, fluid pressure within upper chamber **526** may also be selected or modified to achieve a desired contact pressure between carbon brush **406** and rail surface **408**, indicated as force F**2** in FIG. **6**. Contact pressure for carbon brush **406** can affect current transfer from rail surface **408** to work machine **100**, for example, with too little force F**2** possibly leading to periodic disconnections from rail surface **408** or arcing. Contact pressure for carbon brush **406** can also impact wear on carbon brush **406**, with too much force F**2** possibly leading to premature deterioration of carbon brush **406** due to friction with rail surface **408**. While magnet **410** and sleeve **412** provide a stable and slidable connection with rail surface **408** even over rough surfaces of haul route **101**, the interior of terminal **306** enables independent adjustment of the force F**2** for carbon brush **406** to optimize current transfer and material wear. Pressure values within upper chamber **526** for optimizing these features will be within the knowledge of those skilled in the art based on specific implementations, but in one example the fluid pressure may be about 5 PSI for suitable performance.

(49) Consistent with the principles of the present disclosure, bridging section **540** also provides functionality for terminal **306** similar to an air spring, an airbag, or a fluid suspension between upper section **520** and lower section **500** (and specifically between post **522** and carbon brush **406**). Bladder **546** has a resilient and flexible structure that permits it to be moved in multiple dimensions, such as stretched or contracted along central axis **510**, twisted angularly around central axis **510**, or moved laterally from central axis **510**. Because piston **504** and rod **542** sit loosely within upper chamber **526**, and post bottom **602** sits loosely within lower chamber **512**, bladder **546** further enables lower section **500** to move in multiple dimensions relative to post **522**. For instance, sliding along rail surface **408**, terminal **306** may encounter a dip or a bump in the surface (i.e., a vertical deviation in rail surface **408** along the Z axis). Bridging section **540** enables piston **504**, and lower section **500** generally, to react to the vertical deviation by moving upwardly or downwardly along central axis **510**. In this way, bridging section **540** may act as a shock absorber

for carbon brush **406**. In another example, if terminal **306** encounters a slant in rail surface **408** (i.e., a rotation to a portion of rail surface **408** in the X-Z plane), bladder **546** enables lower section **500** to move laterally along horizontal axis **606** as well as vertically along central axis **510** as necessary to accommodate the slant. As a result, this air-spring functionality for bridging section **540** provides terminal **306** with the ability to adapt to small imperfections in rail surface **408** while maintaining maximum and uniform surface contact between carbon brush **406** and rail surface **408**.
(50) Lower section **500** and bridging section **540** also enable terminal **306** to twist angularly about central axis **510**. Specifically, bearings **516** and rotary seal **518** seal conductive fluid **514** within lower chamber **512** while also permitting piston **504** to rotate about central axis **510** independently from reservoir **506**. The loose association of piston **504** and rod **542** within upper section **520** further frees piston **504** to twist about **501**. Consequently, if carbon brush **406** begins to wear unevenly (e.g., decreasing in thickness at an outer side along horizontal axis **606**), downward force F**1** in some examples will cause a slightly higher force on one side of carbon brush **406**, causing piston **504** and carbon brush **406** to rotate around central axis **510**. This rotation will compensate for and eventually equalize the wear across the contacting surface of carbon brush **406**. Deterioration of carbon brush **406** will therefore be kept uniform across the sliding surface and a maximum contact area with rail surface **408** can be maintained during extended use of terminal **306**. Overall, the free movement of carbon brush **406** due to bridging section **540**, whether vertically, laterally, or angularly, will help maintain a consistent contact area and consistent contact force for carbon brush **406**, increasing contact quality and current flow while decreasing risks of arcing.
(51) Another feature of the current collector **118** arises from the array of terminals **306** configured within frame **302**. As discussed above in reference to FIG. **3**, frame **302** in the illustrated example includes a plurality of terminals **306** for each power rail **108**, such as first inner terminal **306**A-**1**, second inner terminal **306**A-**2**, and second inner terminal **306**A-**2** for inner rail **108**A. Tubes **332** provide pressurized fluid from a common source (not shown) within work machine **100** to each of these terminals. Accordingly, as each of the terminals for a rail may start to wear differently, the fluid pressure provided can help to balance the downward forces F**2** provided across the terminals so that those forces are substantially equal to ensure a constant and consistent contact pressure for each of the terminals. Similarly, as tubes **332** supply fluid pressure from a common source to terminals on other rails, such as middle rail **108**B and outer rail **108**C, uneven wear on terminals between rails can be naturally balanced with distributed fluid pressure.
(52) Although carbon brush **406** may be implemented as a solid structure with a disk-shape, additional versatility for terminal **306** may be achieved by dividing carbon brush **406** into sections. FIG. **7** shows an isometric view of an underside of an alternative implementation of carbon brush **406** and plate **502** with a segmented carbon brush **406**. The structure in FIG. **7**, generally depicted as brush assembly **700**, includes a carbon brush **702** having an overall disk-shape collectively formed from a plurality of sections. In some examples such as illustrated, carbon brush **702** is made of eight sections each having a wedge or pie shape, namely, first wedge **710**, second wedge **720**, third wedge **730**, fourth wedge **740**, fifth wedge **750**, sixth wedge **760**, seventh wedge **770**, and eighth wedge **780**. First wedge **710** is representative and generally includes a first side **712** forming a portion of a circumferential outer surface of carbon brush **702** and a first bottom surface **714** aligned in the X-Y plane and intended to contact rail surface **408** in operation. A first chamfer **716** bridges first side **712** to first bottom surface **714**.
(53) Together, the wedges in carbon brush **702** roughly form a disk as with carbon brush **406** and have a substantially circular bottom surface within the X-Y plane (such as with first bottom surface **714** of first wedge **710**) for contacting rail surface **408** when in operation. A distance across the substantially circular bottom surface and through a center point C is depicted in FIG. **7** as outer diameter OD and, in one example, may be about 3 inches. Carbon brush **702** further includes a central space **706** having a distance across the center point C as an inner diameter ID of, in one

example, about one inch. Shown in FIG. **7** for sixth wedge **760**, wedges are separated by a lateral gap **704**, which in some instances may be about 1/10 of an inch. All of these dimensions are representative only and may differ as needed. Moreover, as shown for first wedge **710**, each section has a wedge angle WA measured from the center point C within the substantially circular bottom surface. In one example, for a carbon brush **702** having eight sections as shown, wedge angle WA is about 45 degrees. Each of the sections of carbon brush **702** is secured to plate **502** by any suitable means, such as adhesion, riveting, bolting, or the like. In one option, a bolt (not shown) through central space **706** along the Z axis clamps the sections against plate **502**. Finally, a shaft **708** extends from plate **502** and may be used to attach brush assembly **700** to piston **504** along central axis **510**. The configuration of FIG. **7** is illustrative only, and the number, shape, and dimensions for the sections (e.g., the number, shape, and/or dimensions of the wedges included in the brush assembly **700**) and support structure of brush assembly **700** may vary for a desired implementation.
(54) The structure and composition of the sections in carbon brush **702** provide added versatility for current collector **118**. For instance, lateral gap **704** between each of the sections, as well as central space **706**, enables air to vent through those areas to help cool carbon brush **702**. In addition, while the material for carbon brush **702** is typically a composite material of graphite blended with various metals, such as copper, silver, or a copper alloy, the use of multiple sections enables a mixing and matching of different compositions to achieve a desired overall performance. In one example, one or more sections of carbon brush **702**, e.g., first wedge **710**, third wedge **730**, fifth wedge **750**, and seventh wedge **770**, have a first composition selected to provide elevated electrical properties, such as low electrical resistance. For that same exemplary carbon brush **702**, one or more other sections, e.g., second wedge **720**, fourth wedge **740**, sixth wedge **760**, and eighth wedge **780**, may have a second composition selected to provide elevated mechanical properties, such as an abrasive property for cleaning rail surface **408** while sliding. Particular compositions for each of the sections may also be chosen based on the environment in which brush assembly **700** is intended to be used, which may impact the wear and duration of carbon brush **702**. The implementation for carbon brush **702** is not limited to these described examples, and other compositions and mixtures between sections are within the knowledge and routine experimentation of those of ordinary skill in the art.
(55) While FIGS. **4-7** are directed to features of terminal **306** and its associated structure, FIG. **8** illustrates the terminals installed within frame **302** and mounted on power rail **108**. FIG. **8** is essentially a rear view of current collector **118** in FIG. **2** looking along the direction of forward travel for work machine **100**, i.e., along the X axis. As indicated in FIG. **8**, inner rail **108**A, middle rail **108**B, and outer rail **108**C and held in place by a plate **802** or bracket that may be elevated above ground by a pole (not shown). Frame **302** forms the support structure for current collector **118**, and sleeve **412** on each terminal **306** glides across a rail surface **408** for separate ones of power rail **108**. As an example, third inner terminal **306**A-**3** is installed within substrate **308** of frame **302** and has third inner sleeve **412**A-**3** contacting inner rail surface **408**A of inner rail **108**A. Although blocked in FIG. **8** by third inner sleeve **412**A-**3**, carbon brush **406**A-**3** is forced against inner rail surface **408**A by fluid pressure injected from tubes **332**. Similar connections exist for **416**B-**3** (within third middle sleeve **412**B-**3**) against middle rail surface **408**B and for **416**C-**3** (within third outer sleeve **412**C-**3**) against outer rail surface **408**C.
(56) FIG. **8** illustrates how angled inner side **310** and angled outer side **312**, together with inner bumper **318** and outer bumper **320**, provide protection and lateral stability for current collector **118** as it slides along power rail **108**. As an example, angled inner side **310** has a first face **804** and angled outer side **312** has a second face **806** each angled to face power rail **108** on the underside of frame **302**. As well, inner bumper **318** has a first wall **808** angled to face inner rail **108**A and a second wall **810** angled to face middle rail **108**B, while outer bumper **320** has a first wall **812** angled to face middle rail **108**B and a second wall **814** angled to face outer rail **108**C. With current collector **118** positioned in contact with power rail **108**, these angled faces and walls help shield the

carbon brushes under the multiple terminals **306** from debris. Moreover, the angled faces and walls can help ensure the lateral positioning for current collector **118**. If trailing arms **116** were to pull current collector **118** horizontally, such as in the direction of the Y axis in FIG. **8**, first face **804** would collide with inner rail **108**A and provide a physical block to resist further movement in that direction. Likewise, second wall **810** on inner bumper **318** would contact middle rail **108**B, and second wall **814** on outer bumper **320** would contact outer rail **108**C, also resisting movement of current collector **118** along the Y axis. Similarly, if trailing arms **116** were to pull current collector **118** in the direction of the −Y axis, second face **806** on angled outer side **312** in some examples would collide with outer rail **108**C, first wall **812** of outer bumper **320** would collide with middle rail **108**B, and first wall **808** on inner bumper **318** would collide with inner rail **108**A, each providing a physical block to resist further movement in the −Y direction. These angled faces and walls permit minor lateral forces to cause current collector **118** to shift or shimmy horizontally on power rail **108** as it slides, but prevent those minor forces from detaching current collector **118**.
(57) Excessive forces from trailing arms **116** that exceed a threshold may cause current collector **118** to become detached and lifted from power rail **108**. In some implementations, trailing arms **116** may be configured such that excessive forces cause current collector **118** to be lifted from power rails **108** only in a vertical direction along the Z axis, such that the carbon brushes remain substantially parallel with the rail surfaces. In other implementations for trailing arms **116**, current collector **118** could be lifted from power rails **108** angularly, such that the carbon brushes are no longer parallel with the rail surfaces. FIG. **9** illustrates an example from this latter implementation, showing how, even with angular detachment of current collector **118**, the angled faces and walls at the underside of frame **302** help avoid incorrect re-engagement between terminals and rails.
(58) In FIG. **9**, current collector **118** is depicted in a condition where an external force from trailing arms **116**, such as through middle contactor lug **226**, has sufficient strength to overcome the adhesion of the multiple magnets to the rails, and current collector **118** is detached and lifted from the rails. In this condition, but for angled inner side **310**, angled outer side **312**, inner bumper **318**, and outer bumper **320**, current collector **118** may be displaced laterally to the point that when current collector **118** drops back to the rails, the carbon brushes become engaged to the wrong rails. For example, if current collector **118** were shifted significantly to the right in FIG. **9**, but for inner bumper **318**, outer bumper **320**, and angled outer side **312**, third middle sleeve **412**B-**3** could land on inner rail surface **408**A and third outer sleeve **412**C-**3** could land on middle rail surface **408**B. This misalignment could lead to problems for work machine **100** if the rails are electrified with different poles. In this circumstance, however, at least first wall **808** in FIG. **9** prevents the rightward shift of current collector **118** by colliding with inner rail **108**A. It will be appreciated that if current collector **118** is lifted farther upwards and rightwards from power rail **108** in FIG. **9**, inner bumper **318** and outer bumper **320** will continue to block a misaligned engagement between terminals and rails. In addition, if current collector **118** were lifted above inner bumper **318** and outer bumper **320** and shifted to the right such that third middle sleeve **412**B-**3** were above inner rail **108**A, outer flange **316** will collide with outer rail **108**C and minimize the risk of engagement of terminals with incorrect rails. Further, should the external forces be such as to continue pulling current collector **118** laterally from power rail **108**, outer bumper **320** will allow frame **302** to slide away from the rails without snagging or otherwise causing a disruptive separation. Similar behavior will follow if the external forces were to pull current collector **118** leftward in FIG. **9** with the angled surfaces of first face **804**, second wall **810**, and second wall **814**, as well as inner flange **314** of frame **302** providing structural protection against improper re-engagement. It will be appreciated that the particular shapes and angles of first face **804**, second face **806**, first wall **808**, second wall **810**, first wall **812**, and second wall **814** can be selected or tuned to achieve desired lateral stability and controlled separation of current collector **118** from power rail **108**.
(59) A method for engaging a slidable electrical contact with a conductor rail is defined by representative steps consistent with the present disclosure in the flowchart of FIG. **10**. For the

method of FIG. **10**, the steps in which the method is described are not intended to be construed as a limitation. Any number of steps can be combined in any order to implement the disclosed method, can be performed in parallel to implement the processes, and in some embodiments, one or more blocks of the process can be omitted entirely. Moreover, the processes can be combined in whole or in part with other methods.

(60) In FIG. **10**, the method **1000** begins with a step **1002** of touching an electrical contact of a slidable terminal on a conductor rail. As indicated above, a carbon brush **406** of a terminal **306** within a current collector **118** may be set on a **408**, as shown in FIGS. **4** and **5**, for example. In step **1004**, the slidable terminal is clamped to the conductor rail using, at least in part, an attractive force between a magnet surrounding a body of the slidable terminal and conductor rail. A comparison of FIGS. **5** and **6** illustrates one arrangement where magnet **410** surrounding body **404** pulls body **404** downward with force F**3** and clamps terminal **306** to rail surface **408**.

(61) Method **1000** continues with step **1006** of applying a downward force along a central axis between the electrical contact and the conductor rail. The method **1000** further includes step **1008** of directing a fluid through an upper section of the terminal and into a region confined by a flexible sleeve, the flexible sleeve coupling the upper section and a lower section of the terminal and being positioned radially around the central axis. FIG. **6** illustrates a representative terminal **306** in which pressurized fluid, typically a gas such as air, is fed through passageway **402** and into cavity **524** and upper chamber **526**. At least upper chamber **526** and middle chamber **554** (FIG. **5**) are confined by bladder **546**, which extends around central axis **510** and bridges post **522** of upper section **520** with reservoir **506** of lower section **500**. In a next step **1010** of method **1000**, pressure for the fluid is adjusted to a first pressure level within the region, which as discussed above leads to force F**2** pressing downwardly along central axis **510** on carbon brush **406** and into rail surface **408**. Finally, step **1012** entails sliding the electrical contact across the conductor rail, which may be seen in at least FIG. **1** as work machine **100** traverses haul route **101** and pulls current collector **118** using trailing arms **116**.

(62) Those of ordinary skill in the field will appreciate that the principles of this disclosure are not limited to the specific examples discussed or illustrated in the figures. For example, while current collector **118** is described having an array of nine terminals in a square grid, any number and arrangement of terminals will suffice. Moreover, while the disclosure describes terminal **306** as having an interplay of pneumatic/hydraulic and magnetic forces, terminal **306** could be implemented without magnet **410** if desired. In addition, the principles disclosed are not limited to implementation on a work machine. Any moving vehicle deriving electrical power from a ground-based conductor rail could benefit from the examples and techniques disclosed and claimed.

INDUSTRIAL APPLICABILITY

(63) The present disclosure provides systems and methods for sliding a current collector across conductor rails to deliver electrical power to a moving work machine, such as a hauler at a mining site. The current collector has an array of terminals with carbon brushes. The terminals have upper sections with conductive posts, lower sections that include reservoirs of liquid metal, and bladders that connect the upper sections with the lower sections. Magnets surround outer shells of the terminals. Fluid above a threshold pressure fed into the bladders holds the upper sections apart from the lower sections to open an electrical path and forces the magnets away from the conductor rails, enabling easy and safe engagement and disengagement with the rails. Fluid below the threshold pressure allows the magnets to clamp the terminals to the conductor rails, lowers the conductive posts into the liquid metal, and urges the carbon brushes against the conductor rails. The bladders provide a fluid suspension distributed across the array of terminals, providing consistent electrical contact between the carbon brushes and the rails and balancing wear of the carbon brushes.

(64) As noted above with respect to FIGS. **1-10**, a terminal **306** for an electric current collector has a lower section **500** including a carbon brush **406**, a reservoir **506**, and a conductive fluid **514**

within the reservoir **506**. The reservoir **506** and the conductive fluid **514** are disposed about a central axis **510**, and the carbon brush **406** and the conductive fluid **514** are electrically connected. An upper section **520** includes a post **522** that is hollow and defines at least an upper chamber **526** along the central axis **510**. A bridging section **540** includes a bladder **546** and an insulative fluid **556** within the bladder **546**. The bladder **546** is pliable, connects the lower section **500** with the upper section **520**, and extends radially around the central axis **510**. The pliable bladder **546** permits movement of the carbon brush **406** with multiple degrees of freedom, which helps ensure stable contact between the carbon brush **406** and a conductive rail and helps achieve even wear for the surface of the carbon brush **406**.

(65) In the examples of the present disclosure, bladder **546** within an array of terminals provide a distributed fluid suspension for a current collector **118**. Acting as a shock absorber or air spring, bladder **546** within a bridging section **540** of a terminal **306** allows for axial, radial, and angular compensation at the interface between carbon brush **406** within a lower section **500** of the terminal **306** and a rail surface **408**. As a result, this fluid suspension can help establish and maintain consistent contact between carbon brush **406** and a rail surface **408**, providing high-quality current transfer for the work machine **100** and enabling balanced wear for the carbon brush **406**. Moreover, with axial extension and compression of terminal **306** controlled by fluid pressure, a current collector **118** can be commanded pneumatically or hydraulically from a source such as work machine **100** to open or close an electrical circuit within current collector **118** and to engage or disengage a magnet clamping force against the **108**, providing versatile and safe operation of the system.

(66) Unless explicitly excluded, the use of the singular to describe a component, structure, or operation does not exclude the use of plural such components, structures, or operations or their equivalents. As used herein, the word “or” refers to any possible permutation of a set of items. For example, the phrase “A, B, or C” refers to at least one of A, B, C, or any combination thereof, such as any of: A; B; C; A and B; A and C; B and C; A, B, and C; or multiple of any item such as A and A; B, B, and C; A, A, B, C, and C; etc.

(67) Terms of approximation are meant to include ranges of values that do not change the function or result of the disclosed structure or process. For instance, the term “about” generally refers to a range of numeric values that one of skill in the art would consider equivalent to the recited numeric value or having the same function or result. Similarly, the antecedent “substantially” means largely, but not wholly, the same form, manner or degree, and the particular element will have a range of configurations as a person of ordinary skill in the art would consider as having the same function or result. As an example, “substantially parallel” need not be exactly 180 degrees but may also encompass slight variations of a few degrees based on the context. While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed machines, systems and methods without departing from the spirit and scope of what is disclosed. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. A terminal for an electric current collector, comprising: a lower section comprising a carbon brush, a reservoir, and a conductive fluid within the reservoir, the reservoir and the conductive fluid being disposed about a central axis of the carbon brush, the carbon brush being electrically connected with the conductive fluid; an upper section comprising a post, the post being hollow and defining at least an upper chamber along the central axis; and a bridging section comprising a bladder and an insulative fluid within the bladder, the bladder being pliable, connecting the lower

section with the upper section, and extending radially around the central axis.
2. The terminal of claim 1, further comprising: a body radially surrounding at least the lower section and the bridging section, the body including a shoulder and a neck, the neck contacting the post of the upper section.
3. The terminal of claim 1, further comprising: a body contacting the post of the upper section; and a magnet contacting the body, the magnet radially surrounding the central axis, the lower section, and the body.
4. The terminal of claim 1, further comprising a cap having a passageway in fluid communication with the upper chamber.
5. The terminal of claim 1, wherein the upper section and the lower section are positioned along the central axis such that the insulative fluid within the bridging section is between the post of the upper section and the conductive fluid of the lower section.
6. The terminal of claim 1, wherein the upper section and the lower section are positioned along the central axis such that at least a portion of the post of the upper section contacts the conductive fluid of the lower section.
7. The terminal of claim 1, further comprising bearings within the lower section configured to enable movement of the carbon brush about the central axis.
8. The terminal of claim 1, further comprising: a piston within the lower section; and a rod secured to the piston and extending through the upper chamber, the rod including a head positioned to limit movement of the upper section away from the lower section along the central axis.
9. The terminal of claim 1, wherein the upper section is attached to the lower section by the bladder.
10. The terminal of claim 1, wherein the carbon brush is divided into sections, pairs of the sections being separated by a lateral gap, at least two of the sections having a different composition.
11. An electric current collector for sliding in a direction along a power rail, comprising: a frame having a substrate with a top surface and an underside; a busbar positioned on the top surface; and a first terminal and a second terminal aligned in a column along the direction and mounted between the top surface and the underside, the busbar electrically connecting the first terminal and the second terminal, wherein the first terminal and the second terminal respectively comprises: a carbon brush, a piston, a reservoir, and a conductive fluid within the reservoir, the carbon brush being aligned and electrically connected with the piston along a central axis, the conductive fluid radially surrounding and contacting at least a portion of the piston along the central axis; an upper section comprising a post, the post being hollow and defining at least an upper chamber configured to loosely receive the piston along the central axis; and a fluid suspension comprising an insulative fluid within a pliable bladder, the fluid suspension connecting the reservoir with the post and enabling movement of the carbon brush along the central axis.
12. The electric current collector of claim 11, further comprising: an angled inner side and an angled outer side on opposite sides of the substrate, the angled inner side having a first face on the underside inclined toward the column, the angled outer side having a second face on the underside inclined toward the column; and an inner bumper on the underside between the column and the angled inner side, the inner bumper having a first wall inclined toward the column.
13. The electric current collector of claim 12, further comprising: an inner flange adjacent the angled inner side; an outer flange adjacent the angled outer side, wherein the inner flange and the outer flange are lower than substrate along the central axis.
14. The electric current collector of claim 11, further comprising tubes connecting the first terminal and the second terminal, wherein the first terminal and the second terminal respectively include a cap having a passageway in fluid communication with the upper chamber.
15. The electric current collector of claim 11, wherein the first terminal and the second terminal respectively further comprise: a body contacting the post; and a magnet contacting the body, the magnet radially surrounding the central axis and the body.

16. A method, comprising: touching an electrical contact of a slidable terminal on a rail surface; clamping the slidable terminal to the rail surface using, at least in part, an attractive force between a magnet surrounding a body of the slidable terminal and the rail surface; and applying a downward force along a central axis between the electrical contact and the rail surface, the applying comprising: directing a fluid through an upper section of the slidable terminal and into a region confined by an airbag, the airbag coupling the upper section and a lower section of the slidable terminal and being positioned radially around the central axis, and adjusting pressure for the fluid to a first pressure level within the region; and sliding the electrical contact across the rail surface.
17. The method of claim 16, wherein the first pressure level positions a post of the upper section within a liquid metal of the lower section.
18. The method of claim 17, further comprising: adjusting the pressure for the fluid to a second pressure level, wherein the second pressure level expands the region confined by the airbag and positions the post apart from the liquid metal.
19. The method of claim 18, wherein the second pressure level generates an upward force on the body greater than the attractive force to overcome the clamping.
20. The method of claim 16, further comprising: moving the electrical contact angularly with bearings about the central axis during the sliding.